

Supplementary Material:

Combining Strain and Ferroelectricity in WSe₂ Transistors for Fast, Low-Power Nonvolatile Logic and Multilevel Memory: A Multiscale Simulation Study

Tanvir M. Mahim*¹ and M. Mosaddequr Rahman¹

¹Department of Electrical and Electronic Engineering, BRAC University, Dhaka 1212, Bangladesh

*Email: tanvir.mahim@bracu.ac.bd

S1 Two-Valley Model of the Strained WSe₂ Valence Band

S1.1 Band structure and statistics

The valence band of monolayer WSe₂ is represented by two parabolic valley systems: the K/K' pair with light hole mass $m_K = 0.36m_0$ and degeneracy $g_K = 2$, and the zone-center Γ valley with heavy mass $m_\Gamma = 2.2m_0$ and $g_\Gamma = 1$. Masses and the lattice constant ($a = 3.32 \text{ \AA}$) are taken from the open Computational 2D Materials Database (C2DB). Hole energies E are measured downward from the K-valley band maximum. The valley edges under biaxial strain ϵ (in percent, tensile positive) are

$$E_K^{\text{edge}} = 0, \quad E_\Gamma^{\text{edge}}(\epsilon) = \Delta E_{\Gamma K}(\epsilon) = 157 \text{ meV} - 341 (\text{meV}/\%) \epsilon, \quad (\text{S1})$$

following the first-principles strain gauge of Afrid et al. (npj 2D Mater. Appl. 2026). Under strong tension E_Γ^{edge} becomes negative, meaning the Γ valley rises above the K maximum and heavy holes dominate. The K edge itself shifts slowly, $dE_K/d\epsilon = -30 \text{ meV}/\%$, which we fold into the flat-band offset of the device solver (Section S4); compression therefore also slightly increases the hole density at fixed bias, consistent with the threshold-voltage drift observed experimentally.

Each 2D parabolic valley contributes a constant density of states $N_v = g_v m_v / (2\pi\hbar^2)$ per spin, and the hole sheet density at hole Fermi level E_F is

$$p_v = 2N_v k_B T \ln \left[1 + \exp \left(\frac{E_F - E_v^{\text{edge}}}{k_B T} \right) \right]. \quad (\text{S2})$$

For a given total density $p = \sum_v p_v$, E_F is found by bisection. At the transport baseline $p = 10^{13} \text{ cm}^{-2}$ the K-valley gas is degenerate with $E_F \approx 33 \text{ meV}$.

S1.2 Areal mass density

The areal mass density used in all phonon scattering rates is computed from the formula mass of WSe₂ (341.3 g/mol) and the unit cell area $\sqrt{3}a^2/2$, giving $\rho_{2D} = 5.93 \times 10^{-6} \text{ kg/m}^2$.

S2 Scattering Rates, Mobility, and Calibration

S2.1 Fermi golden rule rates

All rates are standard 2D forms for parabolic bands. Acoustic deformation potential (ADP) scattering in the elastic, equipartition limit is energy independent:

$$\frac{1}{\tau_{\text{ac}}} = \frac{m_v D_{\text{ac}}^2 k_B T}{\hbar^3 \rho_{2D} v_s^2}, \quad (\text{S3})$$

with $D_{\text{ac}} = 2.3$ eV (K) and 3.1 eV (Γ), and sound velocity $v_s = 3300$ m/s. Zero-order optical deformation potential (ODP) scattering with phonon energy $\hbar\omega_0 = 31$ meV and coupling $D_0 = 4.6 \times 10^8$ eV/cm is

$$\frac{1}{\tau_{\text{op}}(E)} = \frac{D_0^2 m_v}{2\rho_{2D}\omega_0\hbar^2} [N_0 + (N_0 + 1) \Theta(E - \hbar\omega_0)], \quad (\text{S4})$$

where N_0 is the Bose occupation. Intervalley (IV) scattering from a state at hole energy E in valley v into valley v' with edge $E_{v'}^{\text{edge}}$ uses the same zero-order form weighted by the final-state density of states:

$$\frac{1}{\tau_{\text{iv}}(E)} = \frac{D_{\text{iv}}^2 m_{v'} g_{v'}}{2\rho_{2D}\omega_{\text{iv}}\hbar^2} \left[N_{\text{iv}} S(E + \hbar\omega_{\text{iv}} - E_{v'}^{\text{edge}}) + (N_{\text{iv}} + 1) S(E - \hbar\omega_{\text{iv}} - E_{v'}^{\text{edge}}) \right], \quad (\text{S5})$$

with $\hbar\omega_{\text{iv}} = 27$ meV. The sharp onset of the ideal parabolic final-state DOS is replaced by a logistic step

$$S(x) = [1 + \exp(-x/\sigma_{\text{iv}})]^{-1}, \quad \sigma_{\text{iv}} = 90 \text{ meV}, \quad (\text{S6})$$

which represents the combined effect of phonon dispersion, band non-parabolicity, and collision broadening. This broadening is what lets a two-valley model reproduce the gradual strain dependence of the full-band calculation; its influence is quantified below.

Charged impurity (CI) scattering uses a screened Coulomb potential in the momentum relaxation integral

$$\frac{1}{\tau_{\text{CI}}(E)} = \frac{n_{\text{imp}} m_v}{2\pi\hbar^3} \int_0^{2\pi} \left| \frac{e^2}{2\bar{\epsilon}(q + s q_{\text{TF}})} \right|^2 (1 - \cos\theta) d\theta, \quad q = 2k \sin(\theta/2), \quad (\text{S7})$$

with $n_{\text{imp}} = 5 \times 10^{12} \text{ cm}^{-2}$, environment permittivity $\bar{\epsilon} = \epsilon_0(\epsilon_{\text{SiO}_2} + 1)/2$, Thomas-Fermi wavevector q_{TF} of the K-valley gas, and a screening reduction factor $s = 0.8$ accounting for incomplete free-carrier screening of interface centers.

S2.2 Mobility

The momentum relaxation rates add by Matthiessen's rule within each valley. The valley mobility follows the 2D Kubo-Greenwood integral

$$\mu_v = \frac{e}{m_v} \frac{\int_0^\infty \tau_v(E) E (-\partial f / \partial E) dE}{\int_0^\infty f(E) dE}, \quad (\text{S8})$$

and the effective mobility is the density-weighted average $\mu_h = (p_K \mu_K + p_\Gamma \mu_\Gamma)/p$. Figure S1a shows the valley-resolved mobilities versus strain: under compression the light K valley dominates and its intervalley channel freezes out; under strong tension the transport collapses onto the heavy Γ valley. Figure S1b shows that the enhancement is nearly independent of carrier density between 10^{12} and $5 \times 10^{13} \text{ cm}^{-2}$, consistent with the robustness reported by the full-band study.

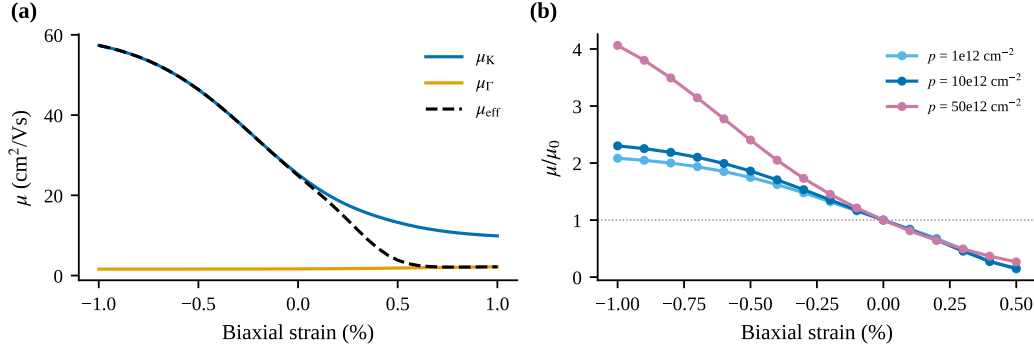


Figure S1: (a) Valley-resolved and effective hole mobilities versus biaxial strain at $p = 10^{13} \text{ cm}^{-2}$. (b) Normalized enhancement for three carrier densities.

S2.3 Calibration and sensitivity

The model has exactly one calibrated coupling, the intervalley deformation potential D_{iv} , plus the broadening σ_{iv} and the screening factor s . They were fixed once by matching three published anchors: the extrinsic baseline mobility ($\approx 25 \text{ cm}^2/\text{Vs}$ at $p = 10^{13} \text{ cm}^{-2}$, $n_{\text{imp}} = 5 \times 10^{12} \text{ cm}^{-2}$, SiO_2 environment), the full-band enhancement curve on the compressive branch, and the strained-FET gauge band at -0.22% . Table S1 compares the calibrated model with the anchors. Figure S2 shows the sensitivity of the enhancement curve to D_{iv} and σ_{iv} : within $\pm 20\%$ of the calibrated values the compressive branch moves by less than the published experimental uncertainty. On the tensile branch beyond $+0.5\%$ the reduced-order model overestimates the mobility loss because the entire hole population transfers to the heavy Γ valley (Figure S1a); the device in the main text never operates there.

Table S1: Validation of the calibrated transport model against published anchors.

Quantity	Model	Published anchor	Source
μ_{h0} (extrinsic)	$24.9 \text{ cm}^2/\text{Vs}$	$\approx 25 \text{ cm}^2/\text{Vs}$	npj 2DM Appl. 10, 57
μ/μ_0 at -0.25%	1.44	1.40	npj 2DM Appl. 10, 57
μ/μ_0 at -0.5%	1.86	1.75	npj 2DM Appl. 10, 57
μ/μ_0 at -1%	2.30	2.37	npj 2DM Appl. 10, 57
μ/μ_0 at -0.22%	1.39	1.70 ± 0.21	ACS Nano 20, 18252

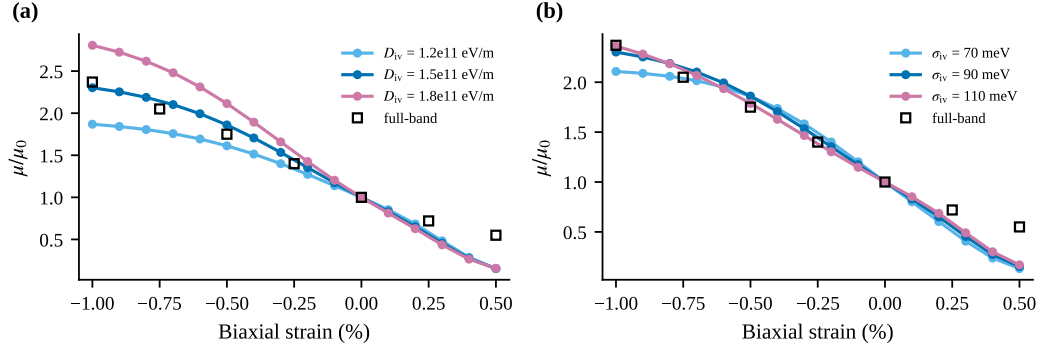


Figure S2: Sensitivity of the strain enhancement to (a) the intervalley coupling D_{iv} and (b) the final-state broadening σ_{iv} . Open squares: full-band results used as calibration anchors.

Table S2: Transport parameters of monolayer WSe₂ used in this work.

Parameter	Symbol	Value	Source
Lattice constant	a	3.32 Å	C2DB
K-valley hole mass	m_K	$0.36 m_0$	C2DB / DFT consensus
Γ -valley hole mass	m_Γ	$2.2 m_0$	DFT consensus
Valley separation	$\Delta E_{\Gamma K}(0)$	157 meV	npj 2DM Appl. 10, 57
Separation gauge	$d\Delta E/d\varepsilon$	341 meV/%	npj 2DM Appl. 10, 57
K-edge gauge	$dE_K/d\varepsilon$	-30 meV/%	ACS Nano 20, 18252
Areal mass density	ρ_{2D}	5.93×10^{-6} kg/m ²	computed
Sound velocity	v_s	3300 m/s	literature
ADP, K valley	$D_{ac,K}$	2.3 eV	literature
ADP, Γ valley	$D_{ac,\Gamma}$	3.1 eV	literature
Optical phonon	$\hbar\omega_0, D_0$	31 meV, 4.6×10^8 eV/cm	literature
IV phonon	$\hbar\omega_{iv}$	27 meV	literature
IV coupling	D_{iv}	1.5×10^{11} eV/m	calibrated
IV broadening	σ_{iv}	90 meV	calibrated
Impurity density	n_{imp}	5×10^{12} cm ⁻²	npj baseline
Screening factor	s	0.8	calibrated

S3 Multidomain CIPS Model

S3.1 Landau coefficients

Each domain follows a second-order Landau expansion $E = \alpha P + \beta P^3$ with $\alpha < 0$, $\beta > 0$. Imposing the measured remanent polarization $P_r = \sqrt{-\alpha/\beta} = 5.5 \mu\text{C}/\text{cm}^2$ and coercive field $E_c = \frac{2}{3\sqrt{3}}|\alpha|^{3/2}\beta^{-1/2} = 0.3$ MV/cm gives

$$|\alpha| = \frac{3\sqrt{3}}{2} \frac{E_c}{P_r}, \quad \beta = \frac{|\alpha|}{P_r^2}. \quad (\text{S9})$$

The non-switching background permittivity is $\epsilon_b = 25$. The coercive field value corresponds to the 2 to 4 V switching onset measured across the 86.5 nm CIPS layer of the experimental MFMIS FeFET.

S3.2 Preisach ensemble

The quasi-static response is an ensemble of 400 to 1600 square hysterons whose coercive fields are drawn from a Gaussian of relative width $\sigma_{E_c} = 22\%$ (clipped to $[0.15, 3] E_c$). Each hysteron contributes $\pm P_r/N$ plus the shared linear background. This spread sets the roundness of the major loop and the shape of all minor loops. Figure S3a shows the major loop for three spreads; the device results of the main text are insensitive to the exact value because the FeFET operates on charge-limited minor loops whose switched fraction is set by the electrostatics, not by the loop shape. Figure S3b shows the hole density during the double sweep of the baseline FeFET, the direct electrostatic counterpart of the transfer curves in main-text Figure 4a.

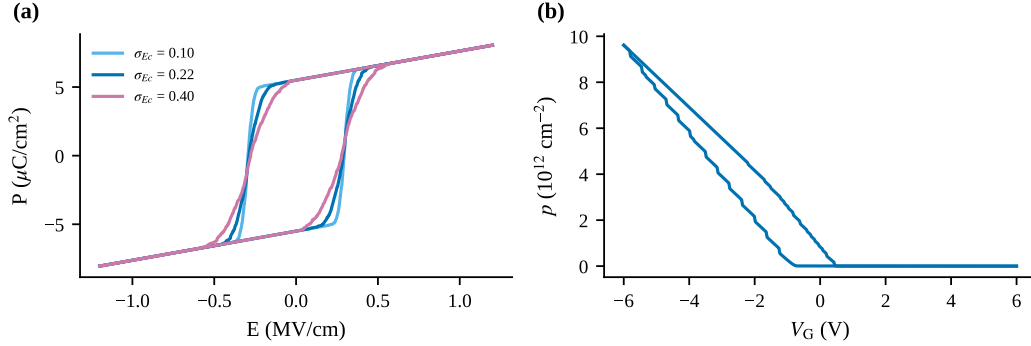


Figure S3: (a) Quasi-static P-E loops for three coercive-field spreads. (b) Channel hole density during the ± 6 V double sweep of the baseline FeFET.

S3.3 Landau-Khalatnikov kinetics

Switching dynamics per domain follow

$$\rho \frac{dP}{dt} = -(\alpha P + \beta P^3) + E(t), \quad (\text{S10})$$

integrated with an explicit scheme on a logarithmic time grid. Domain i uses coefficients scaled to its own coercive field, $\alpha_i = \alpha (E_{c,i}/E_c)$ and $\beta_i = \beta (E_{c,i}/E_c)$, which keeps P_r fixed while distributing switching times. The kinetic coefficient $\rho = 2 \times 10^4 \Omega\text{m}$ is calibrated so that full reversal at $1.5 E_c$ takes about $70 \mu\text{s}$, consistent with the measured $60 \mu\text{s}$ programming pulse of the experimental device, and the resulting field acceleration (main-text Figure 3c) reproduces the observed sub-microsecond partial switching at higher overdrives.

S4 Self-Consistent MFMIS Electrostatics and Retention

S4.1 Stack equations

With the drive variable $x = -(V_G - V_{FB})$ (positive x accumulates holes), zero net charge on the floating gate, and ideal insulators, the stack obeys

$$x = t_{\text{FE}} E_{\text{FE}} + \frac{D}{C_{\text{hBN}}} + \psi_s(p), \quad D = \epsilon_0 \epsilon_b E_{\text{FE}} + P_{\text{sw}}, \quad p = D/e, \quad (\text{S11})$$

where $C_{\text{hBN}} = \epsilon_0 \epsilon_{\text{hBN}} / t_{\text{hBN}}$ with $\epsilon_{\text{hBN}} = 3.5$, $t_{\text{hBN}} = 10 \text{ nm}$, and $\psi_s(p)$ inverts Eq. S2 for the K valley with the strain-dependent flat-band offset $\phi_F(\epsilon) = 0.15 \text{ eV} - 0.030 \epsilon$ set by the doped channel. The h-BN blocks electron injection, so $D \geq 0$; when the solution requires $D < 0$ the channel is depleted, $D = 0$, and the full depolarization field $E_{\text{FE}} = -P_{\text{sw}}/(\epsilon_0 \epsilon_b)$ appears across the ferroelectric.

S4.2 Self-limited hysteron update

Because P_r exceeds the maximum channel charge ($ep \leq 0.5 \mu\text{C}/\text{cm}^2$) by an order of magnitude, forcing full polarization would generate depolarization fields far above E_c and the state would immediately back-switch. The solver therefore alternates the field solution of Eq. S11 with a self-limited update: hysterons whose coercive field is exceeded flip in small batches (four at a time, lowest coercive field first), and the field is re-solved after each batch. The cascade terminates when the internal field is below every remaining coercive field. The stable retained states are partially switched minor-loop states whose polarization matches the available channel charge. This is the mechanism identified for FeFET retention in the classic literature and observed in 2D FeFETs, and it emerges here without any added fitting.

S4.3 Retention proxy and thickness design

Figure S4a shows the retained depolarization field, normalized to E_c , versus CIPS thickness after programming and returning to zero bias. It stays below E_c for all designs, confirming stable retention, but approaches E_c for thick stacks, which explains why very large memory windows come with a retention penalty. Figure S4b and Table S3 give the required program drive and the window for each thickness. The 30 nm design used in the main text is the thickest stack whose full program drive stays within a practical ± 6.6 V while the window (1.3 V) remains compatible with 3 V logic. Note the window depends weakly on the sweep range; the main text quotes 1.24 V for the ± 6 V sweep.

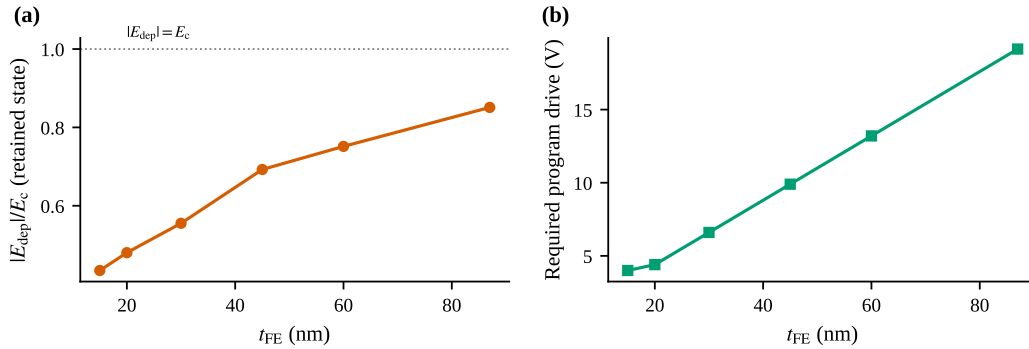


Figure S4: (a) Retained depolarization field over E_c versus CIPS thickness. (b) Program drive used for each thickness.

Table S3: Thickness design of the MFMIS stack (10 nm h-BN throughout).

t_{FE} (nm)	Program drive (V)	Memory window (V)	$ E_{\text{dep}} /E_c$	3 V logic compatible
15	4.0	0.60	0.43	yes
20	4.4	0.79	0.48	yes
30	6.6	1.34	0.56	yes
45	9.9	2.07	0.69	marginal
60	13.2	2.83	0.75	no
87	19.1	4.44	0.85	no

S4.4 Read current

The read current in the linear regime includes the contact resistance explicitly:

$$I_D = \frac{|V_{DS}|}{R_{ch} + 2R_c/W}, \quad R_{ch} = \frac{L}{W e p \mu_h(\epsilon)}, \quad (S12)$$

with $W = 4 \mu\text{m}$, $L = 2 \mu\text{m}$, $V_{DS} = -1 \text{ V}$, and $R_c = 50 \text{ k}\Omega \mu\text{m}$ per contact, chosen inside the span reported for TOS-doped Pd contacts to WSe₂ (1 to 4 $\text{k}\Omega \mu\text{m}$ in optimized processes, 200 to 370 $\text{k}\Omega \mu\text{m}$ in the strained-FET study). This is why the retained read-current gain (1.8 at -1%) is slightly below the mobility ratio (2.3): as the channel improves, the fixed contacts absorb a larger share of the read voltage. Off-state currents are floored at 10^{-13} A to represent the measurement noise floor.

S5 Circuit Models and Nonvolatile Logic Analysis

S5.1 Compact transistor model

Circuit projections use a smooth all-region model. The overdrive is softened by

$$V_{\text{eff}} = 2n\phi_t \ln \left[1 + \exp \left(\frac{V_{GS} - V_T}{2n\phi_t} \right) \right], \quad I_D = \frac{k}{2} V_{\text{eff}}^2 \left[1 - \exp \left(-\frac{V_{DS}}{\max(V_{\text{eff}}, \phi_t)} \right) \right], \quad (S13)$$

with slope factor $n = 1.35$ and $\phi_t = kT/e$. The p-FeFET uses the mirrored form with gain factor $k_p = \mu_h(\epsilon) C_{\text{eff}} W/L$, where C_{eff} is the series capacitance of the CIPS background and the h-BN, and the two stored states carry the thresholds extracted from the device solver: -0.80 V (erased) and $+0.43 \text{ V}$ (programmed) in the gate convention. The n-FET pull-down models the normally-off WSe₂ n-FETs of the experimental latch with $\mu_n = 30 \text{ cm}^2/\text{Vs}$ and $V_{T,n} = 0.6 \text{ V}$.

S5.2 Static analysis

Inverter transfer curves solve $I_p = I_n$ by bisection at each input. The static noise margin of the cross-coupled pair is extracted from the butterfly diagram by the maximum-square method along the diagonal. At $V_{DD} = 3 \text{ V}$ the programmed pair gives SNM = 666 mV. Worst-case static power is the larger of the two off-state leakage products $I_{\text{off}} V_{DD}$; for the resistor-loaded reference it is $V_{DD}^2/R = 0.9 \mu\text{W}$ with $R = 10 \text{ M}\Omega$, matching the experimental configuration.

S5.3 Transient analysis

Node dynamics integrate $C_L dV/dt = I_p - I_n$ with $C_L = 1 \text{ fF}$. For the power-up restore study the supply ramps linearly to 3 V in 0.8 ns while both latch nodes start at zero; the stored complementary FeFET states steer the latch deterministically to its pre-power-loss state (main-text Figure 5c). Figure S5a shows the same restore with a strained pair: -1% compression roughly halves the settling tail. Figure S5b compares worst-case static power of the resistor-loaded and complementary stages on a logarithmic scale. The write path is unchanged by strain: it is limited by CIPS polarization kinetics (Section S3), so write pulses remain in the tens-of-microseconds range at the fields reached in the latch, exactly as measured experimentally.

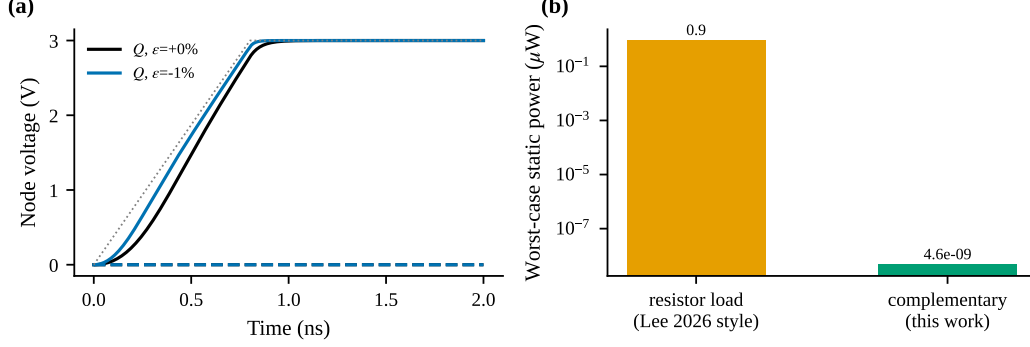


Figure S5: (a) Power-up restore transients for unstrained and -1% strained latches. (b) Worst-case static power per stage: resistor load versus the complementary stage of this work.

S6 Robustness: Interface Traps, Scaling Limits, and Demonstrated Strain Levels

This section quantifies the three factors named in the main text as potential bounds on the conclusions. All results are produced by section [8/8] of `run_all.py` and stored in `results.json`.

S6.1 Worst-case interface trap model

Traps at the $\text{WSe}_2/\text{h-BN}$ interface are modeled as a uniform density D_{it} ($\text{cm}^{-2}\text{eV}^{-1}$) of slow hole traps distributed over a 0.6 eV band of gap states that begins 0.45 eV below the flat-band Fermi offset, so that the interface Fermi level traverses the band as the channel turns on. The captured charge is

$$Q_{\text{it}} = q D_{\text{it}} \text{clip}(\psi_{\text{max}} - \psi_{\text{lo}}, 0, 0.6 \text{ eV}), \quad (\text{S14})$$

where ψ_{max} is the highest surface potential reached since the channel was last fully depleted. The dynamics of this bounding model are chosen as the worst case for a quasi-static sweep: capture is instantaneous whenever the surface potential rises, there is no re-emission while the channel conducts, and complete emission occurs only at the erase extreme (channel fully depleted, trap levels lifted above the Fermi level). Section S6.2 lifts this bounding assumption with finite-rate trap kinetics. Q_{it} enters the displacement balance of the MFMIS solver as semiconductor-side charge, $D = qp + Q_{\text{it}}$, and the charge captured during programming additionally acts as extra Coulomb scatterers in the mobility model ($n_{\text{imp}} \rightarrow n_{\text{imp}} + Q_{\text{it}}/q$) for the retention read. Setting $D_{\text{it}} = 0$ reproduces the trap-free device bit-for-bit.

Figure S6 shows the result of sweeping D_{it} from 10^{10} to $10^{13} \text{ cm}^{-2}\text{eV}^{-1}$. The memory window is remarkably robust: it changes by less than about 6% up to $10^{12} \text{ cm}^{-2}\text{eV}^{-1}$, because trapped charge shifts the forward and reverse threshold voltages together (a rigid shift of up to +0.24 V at 10^{12}) and, by adding to the semiconductor-side charge, lets the ferroelectric ensemble switch slightly deeper rather than eroding the hysteresis. The failure mode is instead the retained current: trapped holes screen the retained polarization, reducing the zero-bias read current from 5.7 μA (trap-free) to 4.8 μA at 3×10^{11} and 2.8 μA at $10^{12} \text{ cm}^{-2}\text{eV}^{-1}$, with the nonvolatile on/off ratio still 2.8×10^7 at 10^{12} ; between 10^{12} and 3×10^{12} the retained state collapses in this bounding model (the finite-rate kinetics of Section S6.2 soften both numbers). The strain gain of the retained current stays between 1.76 and 1.86 for all $D_{\text{it}} \leq 10^{12} \text{ cm}^{-2}\text{eV}^{-1}$, within 4% of the trap-free 1.83, confirming that the strain and trap mechanisms are independent. For context, reported interfaces of 2D channels on h-BN or crystalline insulators reach D_{it} of 5×10^9 to about $10^{11} \text{ cm}^{-2}\text{eV}^{-1}$, typical high- k or native-oxide interfaces 5×10^{11} to 10^{12} , and CVD channels on SiO_2 about $10^{13} \text{ cm}^{-2}\text{eV}^{-1}$.

(Illarionov et al., Nat. Commun. 11, 3385, 2020). The h-BN interlayer assumed here therefore leaves at least an order of magnitude of trap-density margin.

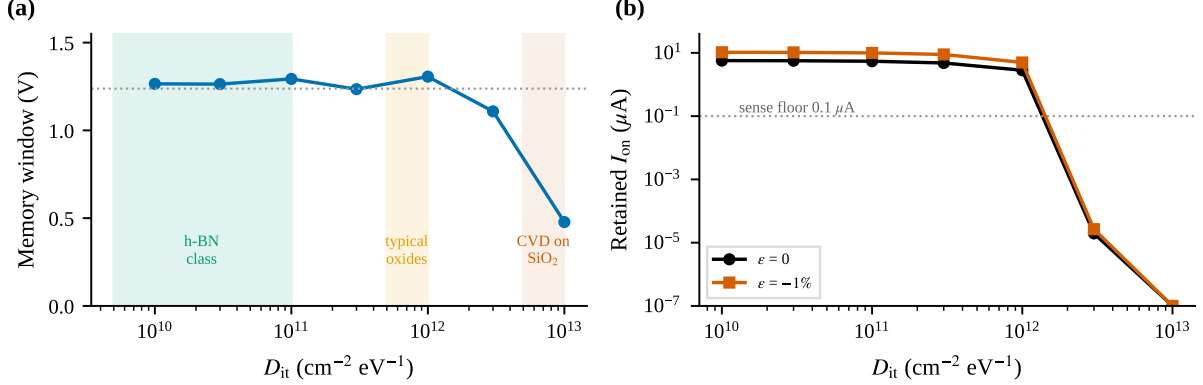


Figure S6: Worst-case interface-trap robustness. (a) Memory window versus trap density; shaded bands mark reported D_{it} ranges for h-BN-class interfaces, typical oxides, and CVD channels on SiO₂. (b) Retained on-current at zero and -1% strain; the logic-compatible regime persists up to 10^{12} cm⁻² eV⁻¹.

S6.2 Finite-rate trap kinetics

To confirm that the conclusions above do not depend on the bounding assumption, the solver also implements energy-resolved Shockley-Read-Hall kinetics. The trap band is discretized into 24 levels $\psi_{t,i}$ whose hole occupancies f_i relax as

$$\frac{df_i}{dt} = c(1 - f_i) - e_i f_i, \quad c = \sigma_p v_{th} \frac{p}{t_{ch}}, \quad e_i = \sigma_p v_{th} \frac{p(\psi_{t,i})}{t_{ch}}, \quad (S15)$$

where p is the instantaneous hole sheet density, $p(\psi_{t,i})$ is the sheet density that would coexist with the Fermi level at the trap level (so the emission rate follows from detailed balance and introduces no parameter beyond the capture cross section), $v_{th} = \sqrt{3k_B T / m_K^*} \approx 2 \times 10^5$ m/s, and $\sigma_p = 10^{-15}$ cm² is a standard interface-trap capture cross section, varied from 10^{-16} to 10^{-14} cm² below. In the steady state Eq. (S15) reproduces the equilibrium Fermi occupancy of each level.

Figure S7a shows the memory window at $D_{it} = 10^{12}$ cm⁻² eV⁻¹ versus the duration of the full double sweep, from 100 μs to 100 s. The window is 1.31 V, constant to better than 10^{-3} V over all six decades, and equally insensitive to the capture cross section across 10^{-16} to 10^{-14} cm². The reason is a self-consistent compensation that we verified directly in the solver: on the return branch the kinetic model holds less trapped charge than the bound (4.4×10^{-4} versus 7.6×10^{-4} C/m² at the turn-off crossing), but the ferroelectric ensemble, which was driven to the same deep minor-loop state at the program extreme by the same total charge, back-switches correspondingly further, leaving the terminal characteristics nearly unchanged. The window conclusion is therefore independent of trap kinetics, not an artifact of the bounding assumption.

Retention is where the bound is genuinely conservative. Figure S7b shows the retained on-current after a 100 μs program pulse as a function of hold time. Trap levels above the retained Fermi level emit their holes on microsecond-to-second time scales, so the read current recovers toward its equilibrium value: at $D_{it} = 10^{12}$ cm⁻² eV⁻¹ the settled current is 3.4 μA (the bound gives 2.8 μA), and at 3×10^{12} , where the bounding model predicts a dead cell, the kinetic model retains 1.7 μA with a nonvolatile on/off ratio of 1.7×10^7 . The strain gain at $D_{it} = 10^{12}$ is 1.85, fully preserved. The window conclusion also extends: at 3×10^{12} the kinetic window is 1.28 V, where the bound gives 1.11 V. At 10^{13} cm⁻² eV⁻¹ both treatments agree that the retained state is lost. The trap-density tolerance of the co-design is therefore at least 3×10^{12}

$\text{cm}^{-2}\text{eV}^{-1}$, more than an order of magnitude above reported h-BN-class interfaces, and the worst-case numbers quoted in the main text are conservative bounds, not best-case estimates.

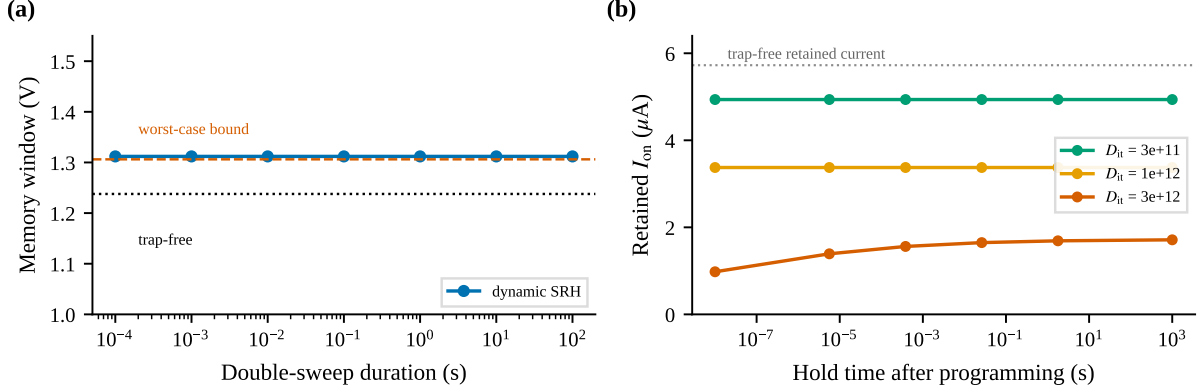


Figure S7: Finite-rate trap kinetics. (a) Memory window at $D_{it} = 10^{12} \text{ cm}^{-2}\text{eV}^{-1}$ versus double-sweep duration: the kinetic result coincides with the quasi-static bound and stays within 6% of the trap-free window over six decades of sweep rate. (b) Retained on-current versus hold time after a $100 \mu\text{s}$ program pulse: emission of trap levels above the retained Fermi level recovers the read current toward its equilibrium value, keeping a working cell even at $3 \times 10^{12} \text{ cm}^{-2}\text{eV}^{-1}$.

S6.3 Short-channel validity of the compact models

The electrostatic scale length of a thin-channel FET, $\lambda = \sqrt{(\epsilon_{ch}/\epsilon_{ins})t_{ch}t_{ins}}$, measures the drain-field penetration into the channel. With the static permittivities of monolayer WSe₂ ($\epsilon_{\parallel} = 15.3$) and h-BN ($\epsilon_{\perp} = 3.76$) from Laturia et al. (npj 2D Mater. Appl. 2, 6, 2018), $t_{ch} = 0.65 \text{ nm}$, and the 10 nm h-BN interlayer, $\lambda \approx 5.1 \text{ nm}$. Long-channel behavior (negligible DIBL and threshold roll-off) requires $L \geq 10\lambda \approx 51 \text{ nm}$. The $2 \mu\text{m}$ device simulated here exceeds this bound by a factor of 40, and the compact circuit models remain valid for any realistic scaled variant down to about 50 nm gate length. Below that, the strain mechanism itself is predicted to survive, with uniaxial compression retaining about half of the biaxial mobility gain.

S6.4 Gains at demonstrated strain levels

Table S4 evaluates every headline metric at the experimentally demonstrated compression (-0.22%), at the intermediate -0.5% , and at the -1% design point. All benefits scale continuously from strain levels available today; none requires new process capability to be observable.

Table S4: Strain-multiplied metrics at demonstrated and projected compression levels (30 nm CIPS, $V_{DD} = 3 \text{ V}$).

Metric	$\epsilon = 0$	-0.22% (demonstrated)	-0.5%	-1%
Mobility gain	1.00	1.39	1.86	2.30
Retained-current gain	1.00	1.29	1.59	1.83
Inverter delay (ps)	50	36	27	22
Energy-delay gain	1.00	1.39	1.85	2.29
Multilevel margin gain	1.00	1.28	1.57	1.80

S7 Code Base and Reproduction Instructions

The complete framework is a pure-Python package (NumPy, SciPy, Matplotlib) organized as follows.

- `params.py`: every physical constant and material parameter, each annotated with its literature or database source.
- `mobility.py`: two-valley Boltzmann transport (Section S2); running it standalone prints the calibration summary.
- `ferro.py`: Preisach ensemble and LK kinetics (Section S3); standalone run prints the loop P_r and switching times.
- `fefet.py`: self-consistent MFMIS solver, double sweeps, program/erase, memory-window extraction (Section S4).
- `circuit.py`: compact models, inverters, butterfly SNM, latch transients (Section S5).
- `run_all.py`: regenerates every figure of the article and this document, plus `results.json` with all headline numbers (including the robustness study of Section S6), in a few minutes on a laptop.

The code base is openly available on GitHub at github.com/Tanvir-Mahmud-Mahim/Process-induced-compressive-strain-with-a-van-der-Waals-ferroelectric-gate-in-a-single-device, and the code, benchmark data, and figure files are permanently archived on Zenodo at DOI: [10.5281/zenodo.22084359](https://doi.org/10.5281/zenodo.22084359). To reproduce: `pip install numpy scipy matplotlib`, then `python3 run_all.py` inside the code directory. All random seeds are fixed, so every figure and number in the article is bit-for-bit reproducible. The strained-transport layer can be recalibrated against future full-band or experimental datasets by editing three parameters (D_{iv} , σ_{iv} , s) in `params.py` and `mobility.py`.